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Suppose V is a vector space over $\mathbb{F}$ with a basis of order n. Then prove "Every basis of V has order n"

Support T is another basis. Since S is independent and T is spanning, $|T| \geq |S|$. The other direction is less trivial, since T might be infitie, and Steinitz does not immediately apply. Instead, we argue as follows: since T is linearly independent, every finite subset of T is independent. Also, S is spanning. So every finite subset of T has order at most $|S|$. So $|T| \leq |S|$. So $|T| = |S|$ *(copy from Part IB - Linear Algebra)*